Weekly check

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. A 7 by 6 cm rectangle loses a 2 by 2 cm corner. Find the remaining area.

2. A parallelogram has base 8, height 3, and slanted side 5 cm. Find its area.

3. Find the surface area of a closed 2 by 3 by 5 cm box.

4. A square pyramid has base side 6 cm and triangular-face height 5 cm. Find surface area.

5. Find $4n + 3$ when $n = 2$.

6. Find $3 \div 8$.

7. Solve $1.5x = 4.80$.

8. Find $\frac{3}{4} \times \frac{4}{5}$.

9. Order $\frac{2}{5}$, 0.45, and $\frac{1}{2}$ from least to greatest.

10. Theo uses a triangle's base 9 cm and height 4 cm. He writes $9 \times 4 = 36$ square cm. What is missing?

11. A rectangle has length 12 cm and width 5 cm. Find area and perimeter.

12. A parallelogram has base 10 cm, height 5 cm, and slanted side 7 cm. Find area and name the unused length.

13. A right triangle has perpendicular legs 20 cm and 5 cm. Find area using either leg as base.

14. A triangle has area 50 square cm and base 20 cm. Find its matching height.

15. A13 cm by 7 cm rectangle loses a3 cm by 2 cm corner. Find remaining area.

16. A triangle and parallelogram share base 10 and height 5. Find each area and their ratio.

17. A closed box measures 2 cm by 5 cm by 10 cm. Find surface area.

18. A cube has side 5 cm. Find surface area and volume; label units.

CHALLENGE / Explain your plan and check the result

C1. A parallelogram has area 72 square cm and base 9 cm. A triangle has the same base and area. Find their perpendicular heights. Explain why the triangle requires twice the height, then find its area if its height is reduced by 3 cm.

C2. A closed $4 \times 5 \times 6$ cm box is wrapped with paper. A label covers 20 square cm already, so no new paper is needed there. Paper costs \$0.02 per square cm. Find the paper area and cost. Explain why volume is not the right quantity.